1. PROTECTED RAW INPUT FROM PWR-SELECT

Mates with PWR-SELECT J301; 15 A path; do not mate live.

C1109 is local high-frequency input decoupling; the clamped bulk bank is not carried by this suspended PCB.

PWR-SELECT C306-C309 provide 27.2 mF plus 660 uF local storage upstream of its load shunt.

2. SYS_4V0 / 12 A SYNCHRONOUS BUCK

4.006 V nominal follows Radxa RK806 guidance. 300 kHz; target loop crossover about 20 kHz.

3. AUX_12V / 8 A FOUR-SWITCH BUCK-BOOST

Revised 12 V / 8 A starting design. Recalculate losses and compensation, then verify by Bode plot and full thermal load before routing.

4. DEDICATED RADIO, NETWORK, AND LOGIC BUCKS

TPS259827L active current limiter; 255 ohm ILIM; 1320 uF local modem bulk; validate weak-signal TX bursts.

TPS22990 provides controlled Wi-Fi startup and PG; it is not a current limiter.

5. LOW-NOISE POINT-OF-LOAD RAILS

TPS62913 uses 2.2 MHz, 470 nF NR/SS and 2.2 uH. Headset LDOs live on Audio-Control; all AKM rails live on AUDIO-8X8.

6. AUX_12V BRANCHES AND AUDIO HANDOFF

DISPLAY_12V: 12 V / 2.5 A harness, simple fuse only. No dedicated monitor eFuse by requirement.

NIGHT_LIGHT_12V: independent 0.25 A fused panel branch; no CM5 or software dependency.

FAN_CPU_12V and FAN_AUX_12V are separate 3 A branches; every fan also has local resettable protection.

AUDIO_12V leaves this board through the controlled inter-board harness. AUDIO-8X8 locally creates +/-15 V, the quiet 5.5 V pre-rail, separate LT3045 ADC/DAC 5 V rails and AKM 3.3 V.

7. SEQUENCING, POWER-GOOD, AND TEST ACCESS

RAW valid -> SYS_4V0, AUX_12V, LOGIC_3V3 and radio pre-regulators start from local UVLO.

SYS_4V0_PG directly enables LOGIC_1V8 and PCIE_1V0; all open-drain PGOOD outputs have 10k LOGIC_3V3 pull-ups.

Wi-Fi and modem final rails default off through 100k pulls; Thermal-IO U1003 asserts them only after checks.

AUDIO_ENABLE controls the AUDIO-8X8 converters; DAC mute stays asserted through rail and clock qualification.

All PGOOD outputs are open drain; control-sheet 10k pull-ups to LOGIC_3V3 and adjacent rail test pads are required.

Component values are Rev A1 starting values; bench loop, thermal, and transfer validation remain ma
DISPLAY_12V is a fused 12 V / 2.5 A harness branch with no dedicated eFuse
The 27.2 mF / 50 V no-blink bank resides on floor-mounted PWR-SELECT
RAW_OUT_LOAD accepts 10.5 V to 30 V after source selection and load telemetry
ProComm
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